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Banking competition in Brazil

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Abstract

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Abstract

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1 Introduction

The Brazilian credit market has undergone significant changes in recent years, driven by new technologies and regulatory reforms that could potentially alter the competitive landscape of the banking sector. This study aims to understand some of the impacts of these changes on the interest rates charged by banks, particularly focusing on the interest rate spread.

Brazilian banking industry is highly concentrated. On December 2024, the five largest banks hold 45.3% total assets of financial institutions, 60.1% of the total loans and 52.8% of total deposits [1]. Even though the concentration is high, it doesn't mean that it is non-competitive, Nakane (2001) tested the level of competitiveness and concluded that Brazilian market doesn't behave as a monopoly/cartel but institutions have some market power.

Santos (2021) used a Cournot model to estimate the effect of different factors on lending rates in Brazil between 2012 and 2016. He concludes that Brazilian rates are higher than other South American countries because of five main factors: IOF tax, high level of risk-free interest rate, higher probability of default, lower recovery rate and more concentrated financial system.

[1] Data from financial institutions accounting reports in https://www.bcb.gov.br/estabilidadefinanceira/balancetesbalancospatrimoniais

2 The model

The model proposed by Ho and Saunders (1981) assumes a market of homogeneous banks working as intermediary dealers. Each bank borrows money from clients to lend it to other clients. Borrowers and lenders arrive randomly on a Poisson distribution, the probability of a new deposit transaction and a new loan transaction are given by:

$$\lambda_D = \alpha - \beta a \tag{2.1}$$

$$\lambda_L = \alpha - \beta b \tag{2.2}$$

Banks can influence the probability of a new deposit or loan by changing it's prices (making it more or less attractive for new clients). The prices for deposits P_D and for loans P_L are:

$$P_D = p + a \tag{2.3}$$

$$P_L = p - b \tag{2.4}$$

where p is the "true" price of the loan or deposit, and a and b are fees that banks can use to increase the probability of a new deposit or loan.

The bank's wealth portfolio (W) is given by $W = Y + I + C$, where Y is the base wealth which is invested in a diversified portfolio, I is the credit inventory and C is the money market position (or short-term net-cash) which is the difference between money market loans borrowings. The expected utility of wealth is given by the following equation:

$$EU(W) = U(W_0) + U'(W_0)r_w W_0 + \frac{1}{2}U''(W_0)(\sigma_I^2 I_0^2 + 2\sigma_{IY} I_0 Y_0 + \sigma_Y^2 Y_0^2) \tag{2.5}$$

The r_w is the expected rate of return on wealth, σ_I^2 is the variance of the credit inventory, σ_Y^2 is the variance of the base wealth and σ_{IY} is the covariance between I and Y. For each new deposit transaction, the initial credit inventory changes by Q and the credit inventory is $I_0 - Q$. The utility of one deposit transaction is:

$$\begin{aligned} EU(W|\textit{one deposit transaction}) = {} & U(W_0) + U'(W_0)aQ + U'(W_0)r_W W_0 \\ & + \frac{1}{2}U''(W_0)(\sigma_I^2 Q^2 + 2\sigma_I^2 QI) \\ & + \frac{1}{2}U''(W_0)(\sigma_I^2 I_0^2 + 2\sigma_{IY} I_0 Y_0 + \sigma_Y^2 Y_0^2) \end{aligned} \tag{2.6}$$

Similarly, for each new loan transaction, the credit inventory is $I_0 + Q$ and the utility of one loan transaction is:

$$\begin{aligned} EU(W|\textit{one loan transaction}) = {} & U(W_0) + U'(W_0)bQ + U'(W_0)r_W W_0 \\ & + \frac{1}{2}U''(W_0)(\sigma_I^2 Q^2 - 2\sigma_I^2 QI) \\ & + \frac{1}{2}U''(W_0)(\sigma_I^2 I_0^2 + 2\sigma_{IY} I_0 Y_0 + \sigma_Y^2 Y_0^2) \end{aligned} \tag{2.7}$$

$$EU(W|a,b) = \lambda_D EU(W|\textit{one deposit transaction}) + \lambda_L EU(W|\textit{one loan transaction}) \tag{2.8}$$

Deriving the equation 2.8 with respect to the fee a to maximize the wealth, we get to:

$$\frac{\partial EU(W|a,b)}{\partial a} = -\beta\left(U'(W_0)aQ + \frac{1}{2}U''(W_0)\sigma_I^2(Q^2+QI)\right) + (\alpha-\beta a)U'(W_0)Q = 0 \tag{2.9}$$

From 2.9:

$$a = \frac{1}{2}\frac{\alpha}{\beta} + \frac{1}{4}\left(-\frac{U''(W_0)}{U'(W_0)}\right)\sigma_I^2(Q+I) \tag{2.10}$$

Doing the same for the fee b:

$$\frac{\partial EU(W|a,b)}{\partial b} = -\beta\left(U'(W_0)bQ + \frac{1}{2}U''(W_0)\sigma_I^2(Q^2-QI)\right) + (\alpha-\beta b)U'(W_0)Q = 0 \tag{2.11}$$

$$b = \frac{1}{2}\frac{\alpha}{\beta} + \frac{1}{4}\left(-\frac{U''(W_0)}{U'(W_0)}\right)\sigma_I^2(Q-I) \tag{2.12}$$

The credit spread is defined by $S = a + b$. Substituting a and b above for S we get:

$$S = \frac{\alpha}{\beta} + \left(-\frac{U''(W_0)}{U'(W_0)}\right)\sigma_I^2 Q \tag{2.13}$$

Defining the Arrow-Pratt coefficient of absolute risk aversion as $R = -\frac{U''(W_0)}{U'(W_0)}$, we can rewrite the credit spread as:

$$S = \frac{\alpha}{\beta} + \frac{1}{2}R\sigma_I^2 Q \tag{2.14}$$

This is the pure spread that will be estimated in the model, in which the first term α/β is the bank's neutral risk spread, which is the part of the risk that is affected by the competition. The second term is a first-order risk adjustment, which depends on the risk aversion (R), the size of the marginal transactions (Q) and the short-term variance of the interest rate (σ_I^2). The model's assumption of homogeneous banks implies that the terms in the credit spread are the same for all banks, even if the inventories (I) and the base wealth (Y) are different.

The empirical model is built based on the theoretical model above but considering three market imperfections: (i) banks also pay implicit expenses through service fees and other costs indirectly involved in the transactions. (ii) the bank's opportunity cost of holding required reserves and (iii) the default risk on loans, for riskier borrowers, the bank may ask for a higher risk premium on the interest rate. The empirical model estimates the bank margins M given by:

$$M = \delta_0 + \delta_1 IR + \delta_2 OR + \delta_3 DP + u \tag{2.15}$$

The intercept δ_0 is the pure spread defined in equation 2.14, IR is the implicit interest expense, OR is the opportunity cost of the required reserves, DP is the default risk premiums on loans and u is the error term.

The equation 2.15 is the first stage of the model used to find the pure spread δ_0 defined on equation 2.14. In the second stage, the regression will be δ_0 estimated for each time t:

$$\delta_t = \gamma_0 + \gamma_1 \sigma_t^2 + \varepsilon_t \tag{2.16}$$

From equation 2.14:

$$\gamma_0 = \frac{\alpha}{\beta} \text{ and } \gamma_1 = \frac{1}{2} RQ$$

Competition is affected by the term γ_0 and γ_1 is the risk aversion (which the model assumes as the same for all banks).

3 Clustering Analysis

The econometric model used in this paper assumes homogeneous behavior among financial institutions. However, it is possible that there are different groups of banks with distinct characteristics and behaviors. Farnè and Vouldis (2021) propose a cluster analysis using the factorial k-means method to identify different business models among banks and suggest this as a way to analyse an "increasingly diverse banking industry landscape". Athey and Imbens (2019) more generally suggest "use cluster membership as a way to organize the sample into types of units that may receive different exposures to treatments."

Our objective is to use the clustering analysis to identify groups of financial institutions with similar characteristics and behaviors, and then the econometric model can be applied to each group separately, allowing for a more precise understanding of the dynamics of the banking industry.

The k-means clustering algorithm is a widely used method for partitioning a dataset into K distinct clusters based on the similarity of the data points. Given a set of observations X_i defined by the balance sheet accounts divided by total assets of the i-th financial institution, the goal is partition the feature space into K subspaces (ATHEY; IMBENS, 2019).

3.1 Data preparation

The data used are monthly financial statement accounts of financial institutions obtained from the Banco Central do Brasil (2025). It covers the period from 2010 to 2024 and includes 206 accounts. Most of those accounts are very specific for some institutions, so the data with more than 30% of missing values were excluded, resulting in 56 accounts. The sample was limited to financial institutions with at least 1 billion reais in credit operations. Since the data is monthly, the value of each variable is the average of the 12 months of the year, institutions that didn't have data for all 12 months were excluded of that year. Then, each variable is divided by the total assets of that institution at that year to make the data comparable across institutions of different sizes. Finally, each variable is standarized with mean zero and standard deviation one.

The database has 3210 observations (financial institution per year) and 56 variables. To exclude outliers of each year, we used the Principal Component Analysis (PCA) method. It was calculated the first 4 principal components for each year and the observations with values higher than 4 standard deviations in any of the first 4 principal components were excluded. This resulted in the final database with 3210 observations.

3.2 Factorial K-means (FKM)

The Factorial K-means was first proposed by Vichi and Kiers (2001) to add dimensionality reduction to the classic K-means.

The k-means algorithm aims to partition the data into K clusters by minimizing the within-cluster sum of squares (WCSS), which is defined as the sum of the squared distances between each data point and the centroid of its assigned cluster. The algorithm starts with an initial set of K centroids, which can be randomly selected from the data points or initialized using a specific method. Then, it iteratively assigns each data point to the nearest centroid and updates the centroids by calculating the mean of the data points in each cluster. This process continues until convergence, which occurs when the assignments no longer change or when a maximum number of iterations is reached.

In the K-means model, given a dataset $X_{N\times P}$ of N banks and P variables and the centroid matrix $Z_{K\times P}$ with K clusters, the objective is to find the the partition $U_{N\times K}$ that minimizes the within clusters sum of squares (WCSS):

$$\min_{U,Z} \sum_{i=1}^{N} \sum_{j=1}^{K} u_{ij} \|x_i - z_j\|^2 \tag{3.1}$$

$u_{i,j}$ is a dummy variable that equals 1 if bank i belonggs to cluster j and 0 otherwise.

FKM adds a projection matrix $V_{P\times d}$ to reduce the original variables P to the factors $d < P$.

Factors are linear combinations of the original variable set subject to the restriction they are orthogonal to each other. Each factor is a vector of loadings for each original variable, loadings range between 1 and -1 and measure the strength and direction of the relationship between the original variable (e.g., Demand Deposits) and the latent dimension (Factor 1). The higher the absolute loading value, the greater the importance of the variable to explain the factor. The optimization problem is to minimize the euclidian distance in the projected space:

$$\begin{gathered}\min_{U,V,Z} \sum_{i=1}^{N} \sum_{j=1}^{K} u_{ij} \|V^T x_i - z_j\|^2 \\ \text{s.t.} V^T V = I_d\end{gathered} \tag{3.2}$$

$V^T x_i$ is the projected point on the factors space.

The FKM algorithm solves this equation using the Alternating Least Squares (ALS) method. It alternates between two steps, it fixes the projection matrix V to find the best U and Z (which is equivalent to the classic K-means) and it fixes U and Z to find the best V. The process is repeated until the WCSS stops decreasing.

The number of clusters and factors is exogenous to the model and there is no single method to help define it. Farnè and Vouldis (2021) use the Hartingan metric witch compares the $WCSS$ using K clusters and $K+1$ clusters.

$$H(k) = \left(\frac{WCSS_k}{WCSS_{k+1}} - 1 \right) \cdot (N - k - 1)$$

The ideal K is defined as the point where the relative reduction in $WCSS$ ceases to be statistically significant enough to justify the loss of degrees of freedom. Hartigan mathematically requires that $WCSS_{k+1} \leq WCSS_k$. It assumes the same space is being partitioned. However, in FKM, the factor matrix changes with each k. When moving to $k+1$, the algorithm creates an entirely new space, which can result in $WCSS_{k+1} > WCSS_k$.

Another method is the Average Silhouette Width (ASW), it compares the average distance of each bank to all banks in the nearest cluster b_i with the average distance of each bank to all the banks in the same cluster a_i. A larger silhouette means the clusters are more separated.

$$s = \sum_{i=1}^{N} \frac{b_i - a_i}{\max(a_i, b_i)}$$

In this study, the method used is the ASW

		Number of factors		
		2	3	4
Clusters	3	0.128		
	4	0.075	0.15	
	5	0.02	0.063	0.097
	6	0.017	0.045	0.052

Table 1 – Average Silhouette Width (ASW)

3.3 Cluster Results

The three factors found were named based on the variables with higher absolute loading values:

Factor 1 (Core deposits) indicates institutions with high liquidity through core deposits, high fixed assets but less diversified operations with low investments and interbank deposits.

Factor 2 (Onlending funding) shows higher credit volume and high Borrowings and onlendings which represents funds raised from other intitutions (development banks, international development agencies).The cost of funds is also higher

Factor 3 (High risk credit) indicates higher risk intitutions more dependent on credit and fee incomes. Low cost of funds may be a consequence of companies that raise funds with equity and securitization strategies.

The clusters were then defined based on the combination of values on each centroid, based on these values we can see the main characteristics that differentiate business models. The differences between the clusters are mostly based on type of funding and revenue sources. The cluster centroids are shown in table 2.

		Factor		
		1	2	3
Cluster	1	0.107	-0.093	-0.234
	2	-0.217	0.199	-0.163
	3	-0.337	-0.019	1.062
	4	0.704	-0.265	-0.128

Table 2 – Cluster Centroids

Cluster 1 (Specialized multi-service banks focused on consumer credit) has the higher proportion of on demand deposits, time deposits, and fixed asset intensity among the four groups, which indicates that this cluster, as well as the second highest concentration of credit operations. This combination characterizes banks with a strong specialization in consumer credit that also possess full access to banking funding sources, structurally distinguishing them from the pure finance companies in Cluster 3.

Cluster 2 (Diversified Conglomerates and Wholesale Players with a Strong Local Presence) comprises the country's five largest banks (Banco do Brasil, Bradesco, Itaú Unibanco, Santander, Caixa Econômica Federal), foreign investment and wholesale banks with substantial local operations (J.P. Morgan, Citibank, Credit Suisse, BNP Paribas, Goldman Sachs, BofA Merrill Lynch, UBS), manufacturer-affiliated captive finance companies (John Deere, Caterpillar, CNH Industrial, Mercedes-Benz, Scania, Toyota, PACCAR, Komatsu), regional development banks, and various niche SCDs/fintechs. In terms of gross averages, this cluster shows the lowest (or near-lowest) values across virtually all line items simultaneously—a pattern consistent with large balance sheets diversified across multiple activities (lending, treasury, foreign exchange, services), which proportionally dilutes each individual ratio. The only notable exception is the highest average for repurchase agreements among the four clusters.

Cluster 3 (Finance Companies) has institutions with operations mostly concentrated in credit, 75% of this cluster has no demand deposits which shows that this cluster has a high concentration of finance companies. This cluster also has high provisions which indicates high-risk portfolios and the highest level of funding via interbank deposits among the four clusters reflecting reliance on the interbank market as a more expensive alternative funding source due to regulatory restrictions on accepting deposits from the general public. Most of the institutions on this cluster are finance companies.

Cluster 4 (Wholesale and Foreign Banks) is the smallest cluster, composed of institutions with a large cash buffer and securities portfolio. It is a strategy typical of Securities holding banks and a large part of this cluster is formed of wholesale foreign banks like Rabobank, Credit Suisse, ING and Deutsche Bank. The cluster profile is dominated by foreign exchange portfolios (on average, 8.2% of the total asset, approximately 15 times the Cluster 2 average) and foreign exchange income (also the highest among the clusters), as well as higher levels of securities, interbank investments, and other receivables, consistent with low local credit intermediation and a focus on treasury and foreign exchange operations.

Figure 1 illustrates the profile of each cluster based on the most discriminating balance sheet accounts, identified from the Factorial K-Means model with the highest loadings in absolute value (the account obligations for borrowings and onlendings under liabilities is divided into three groups: domestic borrowings, domestic on-lendings, and the sum of the liabilities from foreign borrowings and foreign on-lendings). For each account, a z-score was calculated for each individual institution relative to the mean and standard deviation of the full sample, the values represent the distance in standard deviations from the overall sample mean. Subsequently, these standardized scores were aggregated using a simple average within each cluster.

We can see from Figure 1 that Cluster 3 is the one with the most concentrated operations in credit and the largest volume of provisions, indicating a higher risk. Also is more dependent on expensive funding sources like interbank deposits and on-lending. Cluster 4 has high volume of foreign operations indicating the presence of foreign banks. Clusters 1 and 2 have a more balanced profile, but cluster 2 has the highest concentration of domestic intitutional funding and a lower concentration of credit operations than cluster 1, which indicates a more diversified business model while cluster 1 has a more traditional retail bank model.

Figure 2 illustrates the composition of the credit portfolio by risk level for each cluster. Cluster 3 is the one with the highest proportion of high-risk credit operations, Cluster 4 has the portfolio with the lowest risk but has the lower proportion of credit operations over total assets.

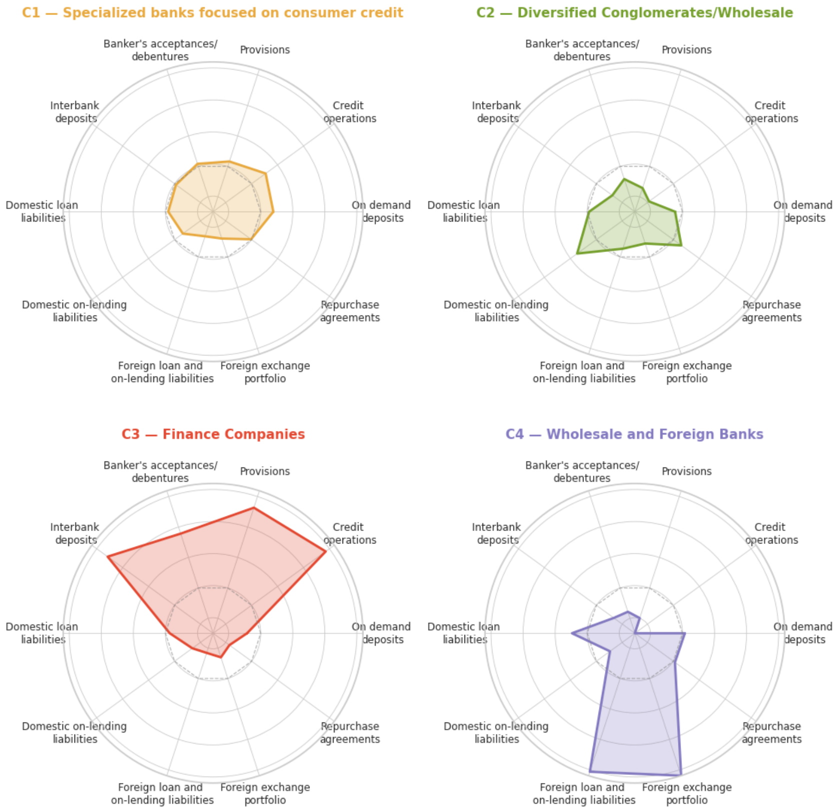

Figure 1 – Comparative profile based on the most discriminating variables

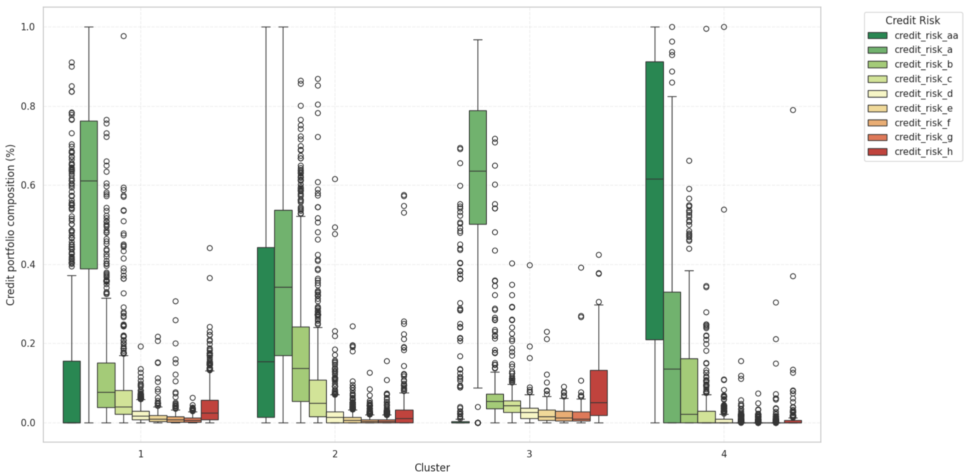

Figure 2 – Composition of Credit Portfolio by Risk Level for each cluster

4 Sample and Data

The empirical model is based on Maudos and Guevara (2004). The variables used and their definitions are shown in Table 3.

4.1 Estimating Lerner index

For the estimation of the Lerner index, we use the same strategy used by Maudos and Guevara (2004). The authors estimate the price as the ratio of total operating revenue to total assets of institution. The cost function is used as natural logarithm of a Cobb Douglas function of the bank outputs (proxied by total assets(TA)), price of labor(w_1), cost of physical capital(w_2) and cost of funds(w_3):

$$\ln C = \alpha_0 + \alpha \ln TA + \beta_1 \ln w_1 + \beta_2 \ln w_2 + \beta_3 \ln w_3 \tag{4.1}$$

For the estimation of the cost function, Pérez de Antón et al. (2003) use a translog cost function, which is a second-order approximation of $\ln C$ adding trend components to capture the time-varying nature of the components, given by the following equation:

$$\begin{aligned} \ln C_i = {} & \alpha_0 + \alpha_1 \ln TA_i + \frac{1}{2}\alpha_2 (\ln TA_i)^2 + \sum_{j=1}^{3} \beta_j \ln w_{ji} + \frac{1}{2}\sum_{j=1}^{3}\sum_{k=1}^{3} \beta_{jk} \ln w_{ji} \ln w_{ki} \\ & + \frac{1}{2}\sum_{j=1}^{3} \gamma_j \ln TA_i \ln w_{ji} + \mu_1 Trend + \mu_2 \frac{1}{2} Trend^2 + \mu_3 Trend \ln TA_i \\ & + \sum_{j=1}^{3} \lambda_j Trend \ln w_{ji} + \ln u_i \end{aligned} \tag{4.2}$$

To estimate the total cost (C) we use the sum of all operating and financial expenses of the institution. The price of labor(w_1) is the administrative expenses divided by the total asset, the price of physical capital(w_2) is the other operating expenses divided by the net fixed assets and the price of funds(w_3) is the total financial expenses (composed by expenses on funding, leasing,securities, loan and on-lending) divided by the total funding (composed by deposits, repurchase agreements, banker's acceptances/debentures and loan and on-lending). The equation was estimated with fixed efects for each bank, in order to control for unobserved characteristics specific to each bank, with robust and clustered standard errors by entity. The usual restrictions of symmetry $(\beta_{jk} = \beta_{kj})$ and homogeneity of degree one in input prices were imposed. The restriction of homogeneity of degree one in input prices is obtained by normalizing the total cost and the prices w_1 and w_2 by the price of funds (w_3).

The cost elasticity on total assets is given by the following equation:

$$\frac{\partial \ln C_{it}}{\partial \ln TA_{it}} = \hat{\alpha} + \hat{\alpha}_k \ln TA_{it} + \hat{\gamma}_1 \ln w_{1,it} + \hat{\gamma}_2 \ln w_{2,it} + \hat{\mu}_3 \, Trend_{it} \tag{4.3}$$

Then the marginal cost is obtained by multiplying the cost elasticity on total assets by the ratio of total cost to total assets:

$$MC_{it} = \left(\frac{\partial \ln C_{it}}{\partial \ln TA_{it}} \right) \times \frac{C_{it}}{TA_{it}} \tag{4.4}$$

Variable	Name	Financial Statement Accounts
nim	Net Interest Margin	$\frac{\text{Operating Revenue} - \text{Funding expenses} - \text{Expenses on borrowings and onlendings}}{\text{Realizable Assets}}$
hhi	HHI Index	$\sum_{i=1}^{n}(marketshare_i)^2$
aoc	Average Operating Costs	$\frac{\text{Operating Costs}}{\text{Total Assets}}$
riskaver	Risk Aversion	$\frac{\text{Total Equity}}{\text{Total Assets}}$
sd	Volatility of market interest rates	
crerisk	Credit Risk	$\frac{\text{provisions for credit operations}}{\text{Total Credit Operations}}$
*sd*crerisk*	Interaction between credit risk and market risk	$sd \times crerisk$
size	Average size of operations	$\ln(\text{Total Credit Operations})$
iip	Implicit interest payments	$\frac{\text{Operating Costs} - \text{Services Income}}{\text{Total Assets}}$
reser	Opportunity costs of bank reserves	$\frac{\text{Cash and Cash Equivalents}}{\text{Total Assets}}$
ef	Quality of manegement	$\frac{\text{Operating Costs}}{\text{Operating Revenue}}$

Table 3 – List of variables used in the empirical model

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Appendix

APPENDIX A – Expected utility of wealth

The expected utility of wealth is given by the equation:

$$E[U(W)] = U(W_0) + U'(W_0)r_w W_0 + \frac{1}{2}U''(W_0)(\sigma_I^2 I_0^2 + 2\sigma_{IY} I_0 Y_0 + \sigma_Y^2 Y_0^2) \tag{A.1}$$

This function based on the assumption that the agents are risk averse, which is mathematically represented as a concave function, which is defined by the condition: $E[U(W)] < U(E[W])$. Ingersoll (1987) shows that

APPENDIX B – Interest rate volatility estimation with IDkA

$IDkA$ (ANBIMA,) is a set of indices that measure the behavior of synthetic portfolios of federal government bonds with a constant maturity. They are calculated by ANBIMA (Brazilian Association of Financial and Capital Markets Entities) using selected vertices of the yield curve of Brazilian government bonds. The indices are calculated for maturities of 3 months, 1 year, 2 years, 3 years, and 5 years.

The index is calculated using the following formula:

$$IDkA_t = IDkA_{t-1} \frac{(r_{t-1}^{n} + 1)^{\frac{n}{252}}}{(r_t^{n-1} + 1)^{\frac{n-1}{252}}} \tag{B.1}$$

where $IDkA_t$ is the index at time t, $IDkA_{t-1}$ is the index at time $t-1$ and r_t^n is the interest rate of the bond with maturity n at time t. Since the government bonds yield curve past data was not available but $IDkA$ database is disclosed by ANBIMA (ANBIMA,). We used the index index data ($IDkA_t$ and $IDkA_{t-1}$) and the yield curve values (r_t)at 11/04/2025 and manipulated the formula to obtain the interest rate from the previous day:

$$r_t^{n-1} = \left(\frac{IDkA_{t-1}}{IDkA_t} (r_{t-1}^{n} + 1)^{\frac{n}{252}} \right)^{\frac{252}{n}} - 1 \tag{B.2}$$

The, we repeated the process for all the available data to obtain the interest rate time series and then calculated the volatility using the standard deviation of the log returns of the interest rates: